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Credit Card Fraud Detection using Machine Learning

Thesis submitted in fulfilment of the requirements for:

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Declaration of Authorship

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|

Dedication

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Acronyms & Abbreviations

AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
TP	True Positive
TN	True Negative
SMOTE	Synthetic Minority Over-sampling Technique
XGBoost	eXtreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
CatBoost	Categorical Boosting
RF	Random Forest
LR	Logistic Regression
PCA	Principal Component Analysis
EDA	Exploratory Data Analysis
API	Application Programming Interface
CV	Cross-Validation
FP	False Positive

Abstract

Credit card fraud represents a growing threat to the global financial system, with annual losses exceeding \$30 billion worldwide (Nilson Report, 2025). Traditional rule-based fraud detection systems, which rely on static thresholds and predefined patterns, are increasingly ineffective against sophisticated, adaptive fraud techniques (Bolton & Hand, 2002). This project addresses this critical challenge by developing and evaluating machine learning models capable of automatically detecting fraudulent credit card transactions in real-time.

Using the publicly available Kaggle Credit Card Fraud Detection dataset (Kaggle, 2023) containing 284,807 transactions with only 492 confirmed fraud cases (0.17% fraud rate) this study implemented and compared five distinct machine learning approaches: Logistic Regression (baseline), Random Forest, AdaBoost, CatBoost, XGBoost, and LightGBM. The dataset's extreme class imbalance presented the primary technical challenge, as a naive model predicting "non-fraud" for every transaction would achieve 99.83% accuracy while detecting zero fraud.

Following rigorous experimentation with stratified train-validation-test splits (70/15/15) and 5-fold cross-validation, XGBoost emerged as the superior model, achieving a test AUC of 0.9778 significantly outperforming all competitors. LightGBM followed with 0.9469, while CatBoost (0.8578), Random Forest (0.8529), and AdaBoost (0.8332) provided moderate performance. Notably, tree-based models handled the severe class imbalance effectively without requiring SMOTE oversampling, relying instead on built-in class weighting mechanisms.

The findings demonstrate that gradient boosting methods, particularly XGBoost, are highly suitable for real-time fraud detection applications, offering both high accuracy and fast inference (<50ms per transaction). This report provides detailed methodology, experimental results, business impact analysis, and deployment recommendations for financial institutions seeking to implement machine learning-based fraud detection systems.

Chapter 1: Introduction and purpose of the study

1.1 Chapter Overview

The rapid growth of digital banking, e-commerce, and online payment systems has fundamentally transformed the financial services landscape. While this digital revolution has brought unprecedented convenience to both consumers and businesses, it has simultaneously introduced new vulnerabilities that fraudulent actors increasingly seek to exploit. Credit card fraud defined as the unauthorised use of another individual's credit card information for the purpose of making purchases or withdrawing funds has emerged as one of the most prevalent and financially damaging forms of financial crime worldwide.

Industry reports indicate that global card fraud losses exceeded USD 30 billion in 2025, with projections suggesting continued growth commensurate with the ongoing expansion of digital transactions. The COVID-19 pandemic served as a significant accelerant of this trend, driving a permanent structural shift towards contactless and online payment modalities that fraudulent actors have been swift to exploit. Traditional fraud detection methodologies principally rule-based systems that flag transactions based on fixed thresholds (e.g., declining any transaction exceeding USD 1,000) are fundamentally constrained in their capacity to adapt to novel and continuously evolving fraud patterns.

These rule-based approaches suffer from two critical limitations: first, they rely on static, manually defined criteria that cannot respond dynamically to the sophisticated and adaptive strategies employed by modern fraudsters; and second, they tend to generate elevated false-positive rates, resulting in the unnecessary declination of legitimate transactions and consequent degradation of the customer experience. There is therefore a compelling need for more intelligent, data-driven detection systems capable of learning from historical transaction data and generalising to previously unseen fraud scenarios.

Machine learning presents a viable and increasingly well-validated alternative to traditional rule-based approaches. By training statistical models on large volumes of labelled transaction data, it is possible to identify complex, non-linear patterns indicative of fraudulent behaviour that would otherwise remain undetected by fixed-rule systems. This study investigates the application of supervised machine learning techniques specifically Logistic Regression and Random Forest classification to the problem of credit card fraud detection using a publicly available, real-world transaction dataset. The proposed pipeline incorporates standard preprocessing procedures, feature scaling, and the application of the Synthetic Minority Over-

sampling Technique (SMOTE) to address the severe class imbalance inherent in fraud detection datasets (Bolton & Hand, 2002).

1.2 Problem Statement

The core challenges this project addresses can be stated as follows: Given a highly imbalanced dataset of credit card transactions where fraudulent cases represent only 0.17% of all transactions, can a machine learning model accurately identify fraudulent transactions while maintaining an acceptable false positive rate?

This problem presents several technical difficulties:

- **Extreme Class Imbalance:** With only 492 fraud cases out of 284,807 total transactions (Kaggle, 2023), any model must learn meaningful patterns from an extremely small minority class. Standard accuracy metrics are misleading a model that predicts "non-fraud" for every transaction achieves 99.83% accuracy but is completely useless for fraud detection (Chawla et al., 2002).
- **Cost Asymmetry:** The costs associated with different types of errors are highly unequal. A false negative (failing to detect a fraudulent transaction) can result in direct financial losses, chargeback fees, regulatory penalties, and customer trust erosion. A false positive (flagging a legitimate transaction as fraudulent) causes customer inconvenience and requires manual review resources but typically carries much lower direct costs (Phua et al., 2010).
- **Concept Drift:** Fraud patterns evolve continuously as fraudsters adapt to detection systems. Any deployed solution must be capable of regular retraining to maintain effectiveness over time (Gama et al., 2014).
- **Data Privacy Constraints:** The dataset features are PCA-transformed, meaning the original transaction attributes are anonymized (Kaggle, 2023). This protects customer privacy but limits direct interpretability of certain feature relationships (Jolliffe, 2002).

1.2 Objectives

This project aims to:

- Conduct comprehensive exploratory data analysis to understand dataset characteristics, feature distributions, and class imbalance
- Implement and compare multiple machine learning models for binary classification (Breiman, 2001; Chen & Guestrin, 2016; Ke et al., 2017)
- Identify the most effective model for credit card fraud detection
- Provide business-oriented recommendations for production deployment (Sahin & Duman, 2011)
- Quantify the potential financial impact of implementing such a system

1.4 Dataset Overview and Exploratory Analysis

1.4.1 Dataset Description

The dataset used in this project was obtained from Kaggle and represents a real-world collection of credit card transactions made by European cardholders during a two-day period in September 2013 (Kaggle, 2023). The data has been anonymized to protect customer confidentiality, with the characteristics shown in Table 1.

Property	Value
Total transactions	284,807
Fraud cases	492 (0.172%)
Non-fraud cases	284,315 (99.828%)
Time span	2 consecutive days (172,792 seconds)
Missing values	0 (complete dataset)

Table 1: Dataset Description 1

1.4.2 Feature Description (Kaggle, 2023):

- V1 through V28: Principal Component Analysis (PCA) transformed features. PCA transforms correlated original features into a set of linearly uncorrelated variables called principal components (Jolliffe, 2002). Due to confidentiality agreements, the original features cannot be disclosed, and the exact interpretation of individual V1-V28 components is not available.
- Time: The seconds elapsed between each transaction and the first transaction in the dataset. This feature captures temporal patterns and potential time-based fraud correlations (Dal Pozzolo et al., 2015).
- Amount: The monetary value of the transaction in euros. This feature was not PCA-transformed and retains its original scale, ranging from 0 to 25,691.16 euros (Kaggle, 2023).
- Class: The binary target variable, where 1 indicates a fraudulent transaction and 0 indicates a legitimate transaction.

1.5 Exploratory Data Analysis Findings

Class Distribution Analysis:

The dataset exhibits severe class imbalance, with non-fraud transactions representing 99.828% of all records (Kaggle, 2023). This imbalance necessitated careful evaluation metrics selection as accuracy alone would be misleading (Chawla et al., 2002). Instead, this study employed ROC-AUC, precision, recall, and F1-score as primary evaluation metrics (Fawcett, 2006).

Temporal Analysis:

Analysis of transaction timing revealed that fraudulent transactions are distributed more evenly across the 48-hour period compared to legitimate transactions (Dal Pozzolo et al., 2015). While legitimate transactions show clear diurnal patterns with lower activity during nighttime hours (approximately 2 AM to 6 AM European time), fraudulent transactions occur at relatively

consistent rates throughout the day. This suggests that fraudsters may operate across multiple time zones or deliberately target periods of lower legitimate activity.

Amount Analysis:

Statistical analysis of transaction amounts revealed notable differences between classes:

Statistic	Non-Fraud	Fraud
Mean	€88.29	€122.21
Median	€22.00	€9.25
Q1 (25th percentile)	€5.65	€1.00
Q3 (75th percentile)	€77.05	€105.89
Maximum	€25,691.16	€2,125.87

Table 2: Transaction Amount Statistics b 1

Interestingly, while the mean fraud amount (€122.21) exceeds the mean non-fraud amount (€88.29), the median fraud amount (€9.25) is substantially lower than the median non-fraud amount (€22.00). This indicates that fraud transactions have a more skewed distribution, with many low-value fraud attempts (possibly testing stolen card validity) and a small number of high-value fraud incidents (Dal Pozzolo et al., 2015).

Correlation Analysis:

The correlation matrix revealed several notable relationships (Kaggle, 2023). V20 and V7 show positive correlation with Amount for legitimate transactions, while V2 and V5 show negative correlation with Amount. No significant correlation was observed between any V1-V28 features and Time, suggesting the PCA transformation successfully decorrelated these features (Jolliffe, 2002). The most important features for fraud detection (as determined by subsequent model feature importance analysis) are V14, V10, V12, V4, and V17, consistent with findings from similar studies (Dal Pozzolo et al., 2015).

Chapter 2: Methodology

2.1. Data Preprocessing

Train/Validation/Test Split:

The dataset was partitioned using stratified random sampling to preserve the fraud ratio across all subsets (Kuhn & Johnson, 2013):

Subset	Size	Fraud Count	Fraud Percentage
Training	70% (~199,365)	~344	0.172%
Validation	15% (~42,721)	~74	0.172%
Test	15% (~42,721)	~74	0.172%

Table 3: Data Split Distribution 1

Feature Scaling:

The Amount feature was standardized using StandardScaler (mean=0, variance=1) to ensure compatibility with distance-based algorithms like Logistic Regression (Kuhn & Johnson, 2013). The Time feature was dropped from model inputs after initial analysis showed no predictive value, reducing potential noise (Dal Pozzolo et al., 2015).

Imbalance Handling Strategy:

Unlike many fraud detection studies that employ SMOTE (Synthetic Minority Over-sampling Technique) (Chawla et al., 2002), this project deliberately opted to evaluate model performance without oversampling. Tree-based models naturally handle class imbalance through their splitting criteria and weighting mechanisms (Breiman, 2001; Chen & Guestrin, 2016). For Logistic Regression, `class_weight='balanced'` was used to adjust penalty for misclassifications (Kuhn & Johnson, 2013). This approach allowed us to assess real-world performance where oversampling might introduce synthetic artifacts (Chawla et al., 2002).

2.2.1 Model Descriptions

Logistic Regression (Baseline):

A linear model that estimates the probability of fraud using a logistic function. Despite its simplicity, Logistic Regression provides an important baseline against which more complex models can be compared. It is highly interpretable and computationally efficient but may struggle with non-linear relationships in the data (Kuhn & Johnson, 2013).

Random Forest:

An ensemble method that constructs multiple decision trees during training and outputs the mode of their predictions. Random Forest reduces overfitting through bagging (bootstrap aggregating) and random feature selection. The model used 100 estimators with a maximum depth of 10 to balance performance and computational efficiency (Breiman, 2001).

AdaBoost (Adaptive Boosting):

A boosting algorithm that sequentially trains weak learners, each focusing on previously misclassified examples. The final prediction is a weighted vote of all learners. AdaBoost is particularly effective at improving the performance of weak classifiers but can be sensitive to noisy data and outliers (Freund & Schapire, 1997).

CatBoost:

A gradient boosting algorithm designed specifically to handle categorical features efficiently. CatBoost uses symmetric trees and ordered boosting to reduce overfitting. While the dataset contains no explicit categorical features, CatBoost's regularization mechanisms still provide benefits for tabular data (Prokhorenkova et al., 2018).

XGBoost (Extreme Gradient Boosting):

The winning model in this study. XGBoost implements regularized gradient boosting with several key advantages (Chen & Guestrin, 2016):

- **Regularization:** L1 and L2 regularization reduce overfitting
- **Tree Pruning:** Depth-first tree building with pruning
- **Handling Missing Values:** Automatically learns direction for missing values
- **Parallel Processing:** Efficient multi-threaded implementation
- **Cross-validation:** Built-in early stopping to prevent overfitting

LightGBM (Light Gradient Boosting Machine):

A gradient boosting framework using tree-based learning with two novel techniques:

- **GOSS (Gradient-based One-Side Sampling):** Retains instances with large gradients
- **EFB (Exclusive Feature Bundling):** Bundles mutually exclusive features
- **Leaf-wise Tree Growth:** Grows tree's leaf-wise rather than level-wise, enabling deeper trees with fewer iterations

2.2.2 Evaluation Metrics

Given the extreme class imbalance, the following metrics were prioritized:

Metric	Definition	Importance
ROC-AUC	Area Under the ROC Curve	Primary metric measures separability ability
Precision	$TP/(TP+FP)$	Cost of false alarms
Recall	$TP/(TP+FN)$	Fraud detection rate
F1-Score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall

Table 4: Evaluation Metrics 1

1. Chapter 3: Experimental Results

3.1 Model Performance Comparison

After extensive experimentation, the following results were obtained:

Model	Validation AUC	Test AUC	Rank
XGBoost	0.9845	0.9778	1
LightGBM (single split)	0.9579	0.9469	2
LightGBM (5-fold CV)	0.9431 (mean)	—	—
CatBoost	0.8578	~0.86	3
Random Forest	0.8529	~0.85	4
AdaBoost	0.8332	~0.83	5

Table 5: Model Performance Comparison 1

XGBoost Detailed Results (Chen & Guestrin, 2016):

- Best validation round: 332 (early stopping)
- Training AUC progression: 0.879 → 0.997 (monotonic improvement)
- Validation AUC peaked at 0.9845 then stabilized
- Test AUC: 0.9778 (excellent separation ability)
- Inference time: <50ms per transaction (real-time capable)

LightGBM 5-Fold Cross-Validation Results (Ke et al., 2017):

Fold	Training AUC	Validation AUC
1	0.9670	0.9946
2	0.977	0.9696
3	0.9814	0.9487
4	0.9704	0.9871
5	0.9728	0.9933
Mean	0.9739	0.9431

Table 6 Validation Results 1

The variance across folds (0.9487 to 0.9946) indicates that LightGBM's performance is somewhat sensitive to the specific training-validation split, suggesting that ensemble methods or more extensive cross-validation may improve stability (Kuhn & Johnson, 2013).

3.2 Feature Importance Analysis

XGBoost Top Features (by importance score):

1. **V14** -Highest importance, strong negative correlation with fraud
2. **V10** -Strong positive correlation with fraudulent transactions
3. **V12** -Moderate positive correlation
4. **V4** - Moderate negative correlation
5. **V17** - Lower but significant importance

Consistent Observations Across Models:

- V14, V10, and V12 consistently ranked among top 5 features across all tree-based models (Breiman, 2001; Chen & Guestrin, 2016; Ke et al., 2017)
- Time and Amount showed minimal predictive value (low importance scores)

- Approximately 8-10 features contributed meaningfully to predictions (Dal Pozzolo et al., 2015)

3.3 Key Findings

Superiority of Gradient Boosting: XGBoost LightGBM significantly outperformed Random Forest, AdaBoost, and Logistic Regression, demonstrating the advantage of boosting over bagging and linear approaches for this dataset (Chen & Guestrin, 2016; Ke et al., 2017).

No SMOTE Required: Tree-based models achieved excellent performance without synthetic oversampling, relying instead on built-in class weighting and algorithmic robustness to imbalance (Breiman, 2001; Chen & Guestrin, 2016; Chawla et al., 2002).

Early Stopping Effectiveness: XGBoost's early stopping at round 332 prevented overfitting, as validation AUC peaked and then plateaued while training AUC continued to improve (Chen & Guestrin, 2016).

Cross-Validation Insights: LightGBM's 5-fold CV revealed performance variance, suggesting that ensemble methods or more robust validation strategies may further improve stability (Kuhn & Johnson, 2013).

Chapter 4: Business Impact and Deployment Recommendation

4.1 Model Recommendation

Based on the experimental results, **XGBoost is recommended for production deployment** due to its superior test AUC (0.9778) (Chen & Guestrin, 2016). LightGBM serves as a viable secondary option for scenarios where training speed is prioritized over marginal performance gains (Ke et al., 2017).

4.2 Expected Performance Metrics (XGBoost)

Metric	Expected Value
Fraud Detection Rate (Recall)	~93-95%
False Positive Rate	<0.1%
Precision	~25-35% (acceptable trade-off) (Sahin & Duman, 2011)
AUC Score	0.9778

Table 7: Expected Performance Metrics 1

4.3 Phased Deployment Strategy (Sahin & Duman, 2011)

Phase 1: Shadow Deployment (2 weeks)

- Deploy XGBoost model alongside existing fraud detection system
- Run in "shadow mode" predictions logged but no action taken
- Compare model predictions with actual outcomes
- Validate performance metrics on live data

Phase 2: Flag and Review (4 weeks)

- Begin flagging transactions predicted as fraudulent for manual review
- Review team investigates flagged transactions

- Collect feedback to refine threshold and identify edge cases
- Monitor false positive rate and adjust thresholds

Phase 3: Auto-Decline (ongoing)

- Auto-decline transactions with fraud probability > 0.85
- Manual review for probabilities between 0.30 and 0.85
- Pass-through for probabilities < 0.30

Phase 4: Maintenance

- Retrain model monthly using last 90 days of labeled transactions
- Monitor for concept drift (performance degradation over time)
- Re-evaluate feature importance periodical

4.4 Financial Impact Analysis (Projected)

Assumptions:

- Average fraud loss per incident: \$500
- Manual review cost per false positive: \$5
- Monthly transaction volume: 1,000,000

Metric	Value
Monthly fraud attempts	~1,700 (0.17%)
Frauds caught (93% recall)	~1,581
Monthly fraud loss saved	\$790,500
False positives (0.09% of non-fraud)	~900
Monthly review cost	\$4,500

Net monthly savings	\$786,000
Estimated annual savings	\$9,432,000

Table 8: Financial Impact Analysis

4.5 Limitations and Future Work

Current Limitations:

- Anonymized features (V1-V28) limit interpretability
- Dataset from 2013 may not reflect current fraud patterns
- Temporal validation not performed (data from only 2 days)
- Precision remains moderate (~25-35%)

Future Work Directions:

1. Hyperparameter optimization for XGBoost using Bayesian methods
2. Ensemble of XGBoost + LightGBM for improved stability
3. SHAP (SHapley Additive exPlanations) for model interpretability
4. Real-time streaming integration with Kafka/AWS Kinesis
5. Automated concept drift detection and retraining pipeline
6. Federated learning for cross-institution model training

4.6 Conclusion

This project successfully demonstrated that XGBoost, a regularized gradient boosting algorithm (Chen & Guestrin, 2016), achieves excellent performance for credit card fraud detection, attaining a test AUC of 0.9778 on a highly imbalanced real-world dataset (Kaggle, 2023). The model significantly outperforms Logistic Regression (Hosmer et al., 2013), Random Forest (Breiman, 2001), AdaBoost (Freund & Schapire, 1997), and CatBoost

(Prokhorenkova et al., 2018), while matching closely with LightGBM (0.9469) (Ke et al., 2017). The approach handles extreme class imbalance effectively without requiring SMOTE oversampling (Chawla et al., 2002), relying instead on built-in class weighting and algorithmic robustness.

The recommended XGBoost model offers substantial business value, with projected annual savings exceeding \$9 million for a mid-sized financial institution (Sahin & Duman, 2011). The phased deployment strategy shadow mode, flag-and-review, and auto-decline provides a practical pathway to production implementation with managed risk (Dal Pozzolo et al., 2015).

As fraud patterns continue to evolve (Gama et al., 2014), regular retraining and monitoring remain essential. Nevertheless, this work establishes a strong foundation for machine learning-based fraud detection that can adapt to new threats while maintaining high accuracy and low false positive rates.

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